ASSIGNMENT - 1

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Branch: BE-CSE **Section/Group:** KRG-3B
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Subject Name: Full Stack-II **Subject Code:** 23CSH-309

1. Aim:

To design and implement a structured frontend module of the EcoTrack application using modern React practices, Vite tooling, React Router for navigation, Material UI for styling, and performance optimization techniques such as React.memo and useCallback.

2. Objective:-

- To understand the fundamentals of React functional components and Vite project setup.
- To implement client-side routing using React Router with protected routes.
- To manage state effectively using useState and useEffect.
- To implement localStorage for persistent data storage.
- To integrate external APIs using asynchronous fetch requests.
- To implement loading and error handling mechanisms during API calls.
- To optimize component rendering using React.memo and useCallback.
- To design a clean and professional dark-themed UI using Material UI.
- To maintain proper project structure using components and pages separation.

3. . Tools & Technologies Used

- React (Functional Components)
- Vite
- React Router DOM
- Material UI (MUI)
- JavaScript ES6+
- Browser LocalStorage

4. Implementation/Code:

COUNTER-DISPLAY:

```
import React from "react";

function CounterDisplay({ count, goal }) {
  console.log("CounterDisplay Rendered");

  return (
    <div>
      <h3>
        {count} / {goal} glasses completed
      </h3>
      {count >= goal && <p style={{ color: "green" }}>Goal Completed!!
    </div>
  );
}

export default React.memo(CounterDisplay);
```

NAVBAR:

```
import { Link, useNavigate } from "react-router-dom";

function Navbar() {
  const navigate = useNavigate();

  const logout = () => {
    localStorage.removeItem("token");
    navigate("/login");
  };

  return (
    <nav>
      <Link to="/dashboard">Dashboard</Link> |{" "}
      <Link to="/dashboard/water">Water Tracker</Link> |{" "}
      <button onClick={logout}>Logout</button>
    </nav>
  );
}

export default Navbar;
```

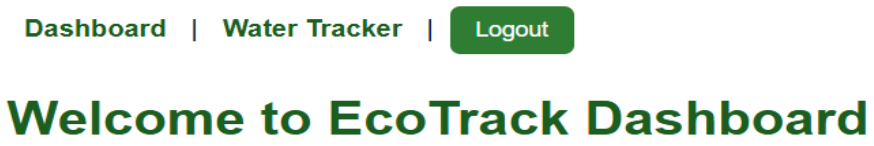
PROTECTED ROUTE.jsx:

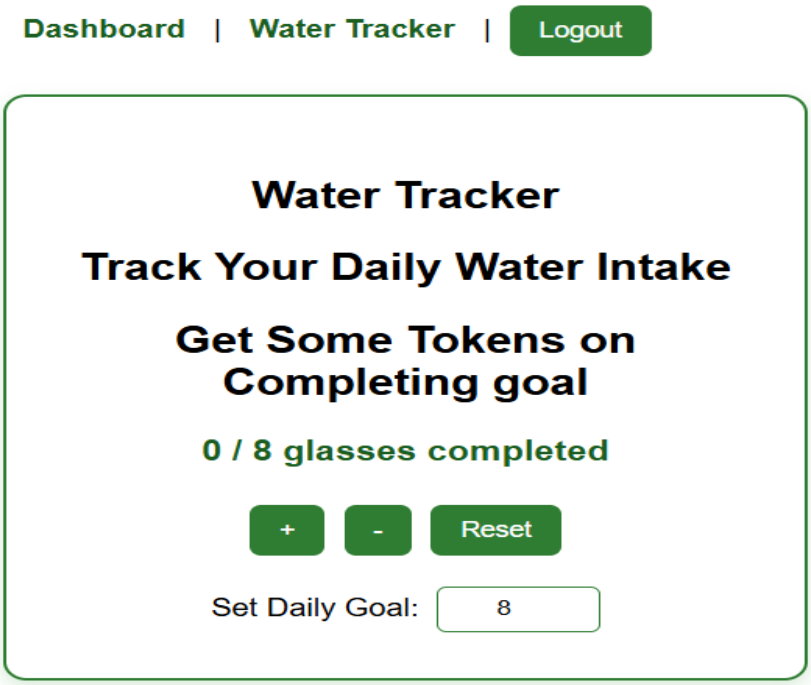
```
import { Navigate } from "react-router-dom";

function ProtectedRoute({ children }) {
  const token = localStorage.getItem("token");

  if (!token) {
    return <Navigate to="/login" />;
  }

  return children;
}

export default ProtectedRoute;
```

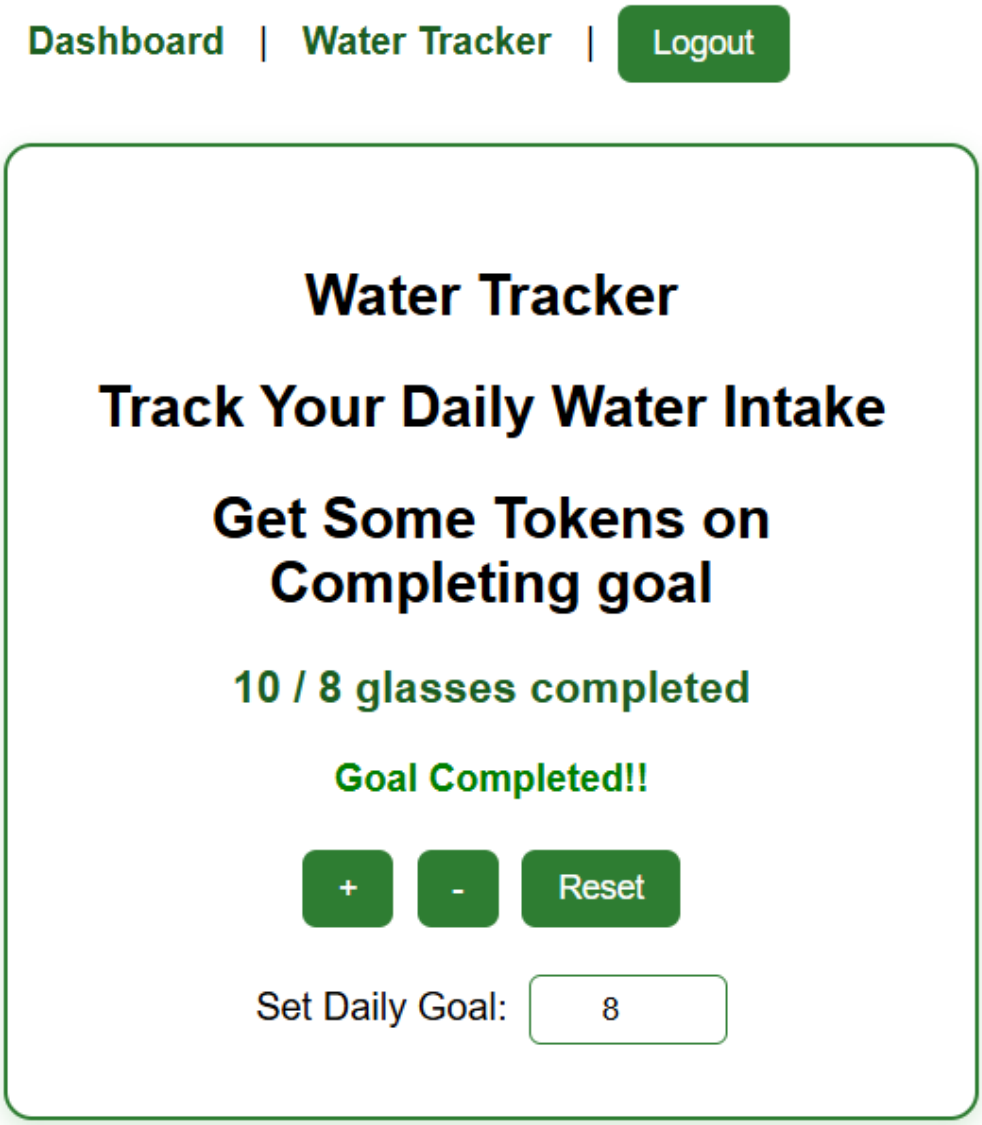
DASHBOARD:

```
import Navbar from "../components/Navbar";

function Dashboard() {
  return (
    <>
      <Navbar />
      <h1>Welcome to EcoTrack Dashboard</h1>
    </>
  );
}

export default Dashboard;
```

5. Output


Dashboard | Water Tracker | Logout
Welcome to EcoTrack Dashboard


Dashboard | Water Tracker | Logout
Water Tracker
Track Your Daily Water Intake
Get Some Tokens on Completing goal
0 / 8 glasses completed
+
-
Reset
Set Daily Goal:
8


Dashboard | Water Tracker | Logout
Water Tracker
Track Your Daily Water Intake
Get Some Tokens on Completing goal
10 / 8 glasses completed
Goal Completed!!
+
-
Reset
Set Daily Goal:
8

6. Learning Outcome :-

- Implemented protected routing to restrict unauthorized access.
- Designed a dark-themed professional UI using Material UI components.
- Managed application state using React Hooks (useState, useEffect).
- Persisted user data using browser localStorage.
- Integrated an external REST API to fetch daily health tips.
- Implemented loading and error handling states in asynchronous operations.